

KASTORIA ARISTOTELIS, GreeceWMO: **166140**Lat: **40.446N**Lon: **21.282E**Elev: **661**StdP: **93.64**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.3	-6.0	-13.6	1.3	-4.8	-11.2	1.5	-3.2	9.6	1.5	8.4	2.0	1.6	300	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	33.9	18.0	32.3	17.7	30.8	17.4	19.7	29.1	18.9	28.4	18.2	27.7	3.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	13.1	21.8	16.0	12.3	20.9	15.2	11.7	20.8	59.1	29.3	56.3	28.5	54.0	27.6	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.8	6.5	5.6	-12.0	36.0	3.4	1.5	-14.5	37.1	-16.5	38.0	-18.4	38.8	-20.8	39.9
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	1.7	4.4	7.5	11.5	15.7	20.2	22.8	23.1	18.6	13.3	7.8	3.5
	DBStd	7.98	4.06	3.52	3.69	3.84	2.71	2.75	2.44	2.02	3.15	3.11	3.62	3.42
	HDD10.0	829	257	158	93	27	1	0	0	0	8	82	203	
	HDD18.3	2512	514	391	334	204	87	13	2	0	35	156	316	459
	CDD10.0	1767	0	1	17	73	178	307	397	407	258	112	16	2
	CDD18.3	410	0	0	0	1	6	70	140	148	43	2	0	0
	CDH23.3	4769	0	0	3	42	124	837	1601	1646	488	29	0	0
	CDH26.7	1866	0	0	0	3	14	280	697	723	144	3	0	0
(14) Wind	WSAvg	1.8	1.6	1.7	2.1	2.1	2.0	2.1	2.3	1.9	1.8	1.4	1.2	1.6
(15) Precipitation	PrecAvg	571	45	45	45	48	48	34	26	27	49	66	70	66
	PrecMax	878	113	120	104	119	103	75	67	88	163	149	157	159
	PrecMin	356	1	4	8	15	7	7	0	1	8	0	20	0
	PrecStd	139	30	33	23	25	22	21	17	18	42	41	34	41
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	18.3	22.4	27.0	28.2	33.2	36.6	36.0	32.1	26.3	20.0	14.8
		MCWB	7.8	9.5	10.7	13.8	15.3	18.4	19.5	17.5	17.1	16.0	11.9	9.7
	2%	DB	11.2	14.9	19.3	24.4	26.2	31.4	34.3	33.9	30.1	23.8	17.5	12.7
		MCWB	6.8	8.3	9.5	12.5	15.3	17.9	18.3	17.6	16.3	14.2	11.1	7.7
	5%	DB	9.3	12.6	17.2	21.5	24.6	29.9	32.5	32.2	28.3	21.9	15.2	11.0
		MCWB	5.8	7.0	8.8	11.6	14.9	17.4	18.0	17.4	16.1	13.6	10.3	7.3
	10%	DB	7.5	10.6	14.8	19.1	22.9	28.2	30.7	30.7	26.4	19.7	13.3	9.5
		MCWB	5.2	6.2	7.9	11.0	14.2	16.8	17.8	17.0	15.7	12.9	9.7	5.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	10.3	12.1	14.3	17.5	20.0	21.9	20.0	21.1	17.9	13.6	11.0
		MCDB	12.3	16.1	20.3	24.5	24.5	28.9	33.4	29.8	26.9	23.7	16.4	12.5
	2%	WB	7.6	8.8	10.4	13.3	16.4	19.0	19.9	19.1	18.0	15.9	12.1	9.4
		MCDB	10.1	13.9	17.5	22.0	23.7	28.4	30.1	28.9	25.0	20.6	15.7	11.5
	5%	WB	6.5	7.7	9.3	12.4	15.6	18.2	19.0	18.4	17.0	14.7	11.2	7.6
		MCDB	8.7	11.7	15.5	19.6	22.9	27.3	29.0	28.2	24.4	18.8	14.2	10.4
	10%	WB	5.4	6.6	8.3	11.4	14.8	17.5	18.3	17.8	16.4	13.9	10.3	6.2
		MCDB	7.3	10.1	14.2	17.8	21.5	26.4	28.1	27.6	23.3	17.6	12.9	8.7
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.9	10.3	11.7	12.1	13.4	14.1	14.8	12.3	10.3	9.1	9.2
		MCDBR	11.2	13.8	15.6	16.2	15.5	16.4	17.4	17.3	16.4	14.8	12.8	11.9
	5% WB	MCWBR	7.2	7.9	7.4	6.6	6.2	5.3	5.2	4.7	5.3	6.5	7.3	7.8
		MCWBR	9.6	12.4	14.0	14.5	14.2	15.2	15.5	14.7	13.2	11.3	10.5	10.0
(40) Clear-Sky Solar Irradiance	taub		0.326	0.356	0.404	0.465	0.461	0.439	0.447	0.459	0.440	0.396	0.355	0.327
		taud	2.385	2.277	2.161	2.000	2.069	2.183	2.161	2.124	2.144	2.259	2.339	2.374
	Ebn at Noon		827	847	840	811	823	841	829	807	793	789	778	790
		Edh at Noon	84	108	136	171	163	146	148	149	137	109	88	79
(44) All-Sky Solar Radiation	RadAvg		1.83	2.60	3.70	4.75	5.74	6.63	6.94	6.15	4.53	3.19	2.12	1.62
	RadStd		0.25	0.35	0.43	0.52	0.39	0.49	0.38	0.38	0.37	0.34	0.21	0.28

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	+0.40	N/A	N/A	+0.71	N/A	N/A	N/A	-100	+100	N/A	N/A

Nomenclature: See separate page